GABRIEL SIMBA MUTAVI

JUNIOR SOFTWARE ENGINEER | FULL-STACK DEVELOPER

Location: Machakos, Kenya

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1. PROFESSIONAL SUMMARY

Junior Software Engineer and Full-Stack Developer with software engineering training from Moringa School and a Diploma in Mechanical Engineering (Production Option) from The Nairobi National Polytechnic. Skilled in Python, Flask, JavaScript, React, SQL, REST APIs, SQLAlchemy, HTML, CSS, Git/GitHub and data management. Experienced in developing database-driven applications, integrating frontend interfaces with backend APIs, maintaining records and organising production data. Also brings practical experience in biopesticide production, moringa-based natural products and chemical product handling through work at Semi Arid Regions and Environmental Solutions NGO. Combines technical knowledge with analytical thinking, problem-solving, teamwork and a willingness to learn.

2. TECHNICAL AND SOFTWARE SKILLS

- **Programming Languages:** Python, JavaScript
- **Frontend Development:** React, HTML, CSS
- **Backend Development:** Flask
- **Databases:** SQL, SQLAlchemy, database design and management
- **Data Management:** Data organisation, record-keeping, data entry, updating records and maintaining accurate production data
- **API Development:** REST APIs and API integration
- **Authentication:** JWT authentication
- **Version Control:** Git and GitHub
- **Payment Integration:** M-Pesa Daraja API integration in a test environment
- **Other Technical Skills:** Application testing, debugging, troubleshooting and frontend-backend integration

3. SOFTWARE ENGINEERING EDUCATION

Software Engineering

Moringa School

February 2026 – September 2026

- Trained in full-stack software development, including frontend and backend development.
- Gained practical experience with Python, Flask, JavaScript, React and SQL databases.
- Learned to develop and integrate REST APIs, implement authentication and manage databases.
- Used Git and GitHub for version control and collaborative software development.
- Worked on software projects involving application development, debugging and testing.

4. SOFTWARE DEVELOPMENT PROJECT

Fleet-Pesa — Fleet Management and Payment Platform

Project Overview:

Fleet-Pesa is a fleet management and payment platform designed to help fleet owners manage vehicles, drivers, remittances and payment records while reducing errors associated with manual cash handling and record-keeping.

Responsibilities and Contributions:

- Developed and integrated frontend and backend components using React and Flask.
- Worked on REST API endpoints to support application functionality.
- Used SQL databases and SQLAlchemy for data management.
- Implemented JWT-based authentication workflows.
- Worked on vehicle and driver assignment functionality.
- Contributed to remittance tracking, payment records and payment history.
- Contributed to fare-payment functionality and M-Pesa Daraja STK Push integration in a test environment.
- Used Git and GitHub for version control and project collaboration.
- Assisted with debugging, testing and resolving application integration issues.

Technologies Used: Python, Flask, JavaScript, React, SQL, SQLAlchemy, REST APIs, JWT, Git and GitHub.

5. PROFESSIONAL WORK EXPERIENCE

Semi Arid Regions and Environmental Solutions NGO

Production Assistant — Natural Products, Biopesticides and Data Management

April 2025 – January 2026

- Participated in the production of biopesticides and other natural products.
- Worked with moringa-based products and supported their preparation and processing.
- Assisted with the preparation and handling of chemical products in accordance with workplace procedures.
- Managed and maintained production records and related data to support accurate record-keeping and reporting.
- Organised and updated data relating to production activities and product information.
- Supported product preparation, packaging and general production activities.
- Maintained attention to product quality, cleanliness and safe handling practices.
- Collaborated with colleagues to support efficient production and data management processes.

6. MECHANICAL ENGINEERING EDUCATION

Diploma in Mechanical Engineering — Production Option

The Nairobi National Polytechnic

September 2021 – April 2025

Relevant areas of study included:

- Production engineering and manufacturing processes.
- Engineering materials and workshop practices.
- Industrial operations and maintenance principles.
- Engineering problem-solving and technical procedures.
- Workplace safety and quality awareness.

7. SECONDARY EDUCATION

Kenya Certificate of Secondary Education (KCSE)

Mumbuni Boys High School

2017 – 2020

8. INDUSTRIAL ATTACHMENT EXPERIENCE

Simba Cement

Industrial Attachment

- Gained exposure to industrial production and cement manufacturing processes.
- Developed an understanding of industrial equipment, workplace procedures and safety practices.
- Observed the importance of teamwork, quality standards and efficient production operations.
- Strengthened practical technical knowledge and problem-solving skills.

Ruiru Steels

Industrial Attachment

- Gained exposure to steel manufacturing and production processes.
- Developed an understanding of industrial equipment and production workflows.
- Learned the importance of workplace safety, teamwork and adherence to operational procedures.
- Strengthened technical awareness and adaptability in an industrial environment.

9. CORE COMPETENCIES

- Problem-solving and analytical thinking
- Data management and record-keeping
- Teamwork and collaboration
- Communication and interpersonal skills
- Critical thinking
- Attention to detail
- Adaptability and willingness to learn
- Time management and organisation
- Debugging and troubleshooting
- Commitment to continuous improvement

10. LANGUAGES

- **English:** Good working proficiency
- **Kiswahili:** Good working proficiency

11. CAREER OBJECTIVE

To build a successful career in software engineering by developing reliable, efficient and user-friendly software solutions. I am particularly interested in full-stack development, backend systems, API integration, data management, fintech and business applications. I aim to apply my technical training, analytical skills and practical work experience to contribute to an organisation's goals while continuing to grow professionally.

12. REFERENCES

1. Lorenah Mbogo

Position: Software Engineering Instructor

Institution: Moringa School

Email: lorenah.mbogo@moringaschool.com

2. Charles Kisima Kimani

Position: Former Supervisor

Company: Semi Arid Regions and Environmental Solutions

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